

DNA Repair and Genome Stability

Sérgio F genetic report
2 of October 2026 (local time)

Name	Sérgio F
Focus	longevity and immortality (84 aging mechanisms)
Data	Crude DNA (chip) + local imputation (only DR2 ≥ 0.8)
Content	16 Curated SNPs · 17 loci of GWAS · 3 polygenic scores · 35 679 pathogenic variants of ClinVar in 63 genes

What is: Sérgio F variants in the pathways that repair DNA (base excision, nucleotide excision, homologous recombination, repair of pairing errors, p53 response) and genetic markers of genome instability (loss in mosaic of chromosome Y), plus a search for rare pathogenic variants in 63 repair genes and hereditary cancer.
Privacy: the analysis was done locally; the genotypes never left the computer. Public databases were only sent SNP identifiers.

Important: genetic predispositions, not a diagnosis. The conclusions come only from the genes (without analysis or exams). All practical ideas are suggestion (not applied), to decide with the doctor.

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1. Abstract: favourable and unfavourable

Favourable	Unfavourable
• Common repair routes without lower efficiency alleles: XRCC1 Arg399/Arg399 (CC), XRCC3 Thr241/Thr241 (GG), ERCC2/XPD Lys751/Lys751 (TT) and Asp312/Asp312 (CC), XPC Lys939/Lys939 (TT) and MLH1 –93 GG. • Non carrier of the rare risk variants tested: TP53 rs78378222, BRCA2 K3326*, CHEK2 I157T. • Y mosaic loss: Score at the 86th percentile of Y dosage (less predicted loss), an indirect signal of good response to DNA damage in the blood. • Prostate cancer: Polygenic score at the 1st percentile. • ClinVar: no plausible pathogenic variant in the 63 genes of the panel; the only hit (ATM in homozygous) is artefact.	• TP53 Arg72/Arg72 (commitment): more apoptosis and less senescence; in an elderly study, Pro72 was associated with more survival despite more cancer. • OGG1 Ser326/Ser326, PARP1 Val762/Val762, APEX1 and MDM2 SNP309 in heterozygosity: in the average of the population (no advantage or weakness). • 2 of the 17 loci of Y loss with more unfavourable alleles than the average (NREP, NPAT). • Untested in the chip: most of the rare variants of BRCA1/2, Lynch syndrome, ATM, PALB2 and CHEK2 1100delC. Only clinical sequencing excludes them.

Reading for immortality: DNA repair appears as a strong relative point: no common allele of minor repair in the main pathways, instability scores (loss of Y) and favourable prostate, and without rare plausible variants. The only ambiguous signal is TP53 Arg72/Arg72, which exchanges less cancer for less protective senescence. The strategy for immortality is to reduce the entry of damage (UV, burned food fertilizers, smoke, radiation) and monitor somatic mutations (traits and, if there is a family history, clinical sequencing).

2. Method, chain and trust

- **Genotypes:** 19 read directly on the chip (GRCh38, strand +); 12 locally imputed (Beagle 5.5, full genome reference panel 30x, only DR2 ≥ 0.8); 2 without data ("SNP not found in the data").
- **Strand:** All alleles were aligned to the GRCh38 strand + and verified against alleles and frequencies of the reference panel. 0 alleles of the literature were in the strand coding – and were converted (complement); 5 palindromic SNPs (A/T or C/G) were resolved by the allelic frequency; the non-resolvable palindromic SNPs were out of the scores.
- **Classification:** compares the copies of the effect allele with the average of the reference population ($2 \times$ frequency). Difference ≥ 0.5 copies $\rightarrow$ Favourable or Unfavourable; lower $\rightarrow$ Typical. Not having a rare variant of risk, or a common variant of strong effect, counts as Favourable. An allele with opposite effects (e.g. longer telomeres but higher risk of cancer) appears as Commitment in both lists.
- **Trust:** chip > imputation (DR2 ≥ 0.8 ; estimated accuracy of ~95–99% in common variants, lower in rare ones). Evidence: Strong (replicated in large/functional GWAS), Moderate, Weak (small or inconsistent studies).
- **Polygenic scores:** weighted sum of alleles (published coefficients); percentile calculated against the reference population with exactly the same variants.
- **Sources of interpretation:** dbSNP/Ensembl, ClinVar, GWAS Catalog, PGS Catalog and articles cited in the text.

3. Polygenic scores

Score	Percentile	Variants used	Interpretation	Note
Y-chromosome dosage (less mosaic loss; 14 loci) G_MLOY · calculated in this report	86	14 of 14 (100%)	Favourable	High percentile = more predicted Y dosage = less Y loss in blood cells. Y loss loci are mainly genes of DNA damage and cell cycle.
Colorectal cancer (76 variants) PGS000055 · previous genomic imputation (same method)	56	74 of 76 (97%)	Typical	Typical.
Prostate cancer (269 variants) PGS000662 · previous genomic imputation (same method)	1	221 of 269 (82%)	Favourable	Very low percentile : less polygenic risk.

the 50th percentile : median of the reference population. For risk scores, high percentile = higher risk; for "pro" scores (e.g. longer telomeres, longevity), high percentile = more of the trait. Typical = percentile between 30 and 70. The scores explain only a small part of the variation and are not diagnostic.

4. Favourable

Gene/score	SNP	Genotype	Source · evidence	Effect
XRCC1	rs25487 Arg399Gln (base excision repair)	CC 0× T	chip Weak	Without the T-risk allele. The Gln399 allele is associated, in functional studies, with lower efficiency of simple chain breaks repair and base excision; in meta-analyzes the increase in cancer risk is small and inconsistent. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
XRCC3	rs861539 Thr241Met (homological recombination)	GG 0× A	chip Weak	Without the risk A allele. The Met241 allele is associated with more fertilizers in DNA and a slightly higher risk of some cancers (bladder, breast) in meta-analyzes. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
ERCC2	rs13181 Lys751Gln (XPD, nucleotide excision)	TT 0× G	chip Weak	Without the risk allele G. The Gln751 allele is associated with lower nucleotide excision repair capacity (UV lesions and fertilizers) in some studies; small effect. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
ERCC2	rs1799793 Asp312Asn (XPD)	CC 0× T	chip Weak	Without the T-risk allele. The Asn312 allele is associated with lower repair of fertilizers in some functional studies; small effect. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
XPC	rs2228001 Lys939Gln (recognition of lesions)	TT 0× G	chip Weak	Without the risk allele G. The Gln939 allele is associated with less repair of large lesions and a slightly higher risk of bladder cancer. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
TP53	rs1042522 Pro72Arg	CC 2× C	imputation (DR2=0.97) Moderate	Arg72: p53 induces apoptosis more efficiently, eliminating cells with damaged DNA (protection against cancer). Mechanisms: Genomic instability; Cell Senescence; Somatic Mutation Accumulation
TP53	rs78378222 polyadenylation signal (AATAAA>AATACA)	TT 0× G	chip Moderate	Without the risk allele G. Rare allele that reduces the processing of p53 mRNA; it is associated with basal cell carcinoma, glioma, prostate cancer and more mosaic loss of Y chromosome. Mechanisms: Genomic instability; Accumulation of Somatic Mutations Somatic / Tissue Mosaicism
BRCA2	rs11571833 K3326*	AA 0× T	chip Moderate	Without the T-risk allele. A rare variant of late truncation, non-pathogenic for classical hereditary cancer, but with a higher risk of 2-2.5 of squamous lung cancer and a modest risk of other cancers. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability
CHEK2	rs17879961 I157T	AA 0× G	chip Moderate	Without the risk allele G. A rare variant of moderate risk (prostate, breast, kidney, chronic lymphocytic leukemia) and clonal hematopoiesis. Mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability; Clonal hematopoiesis (CHIP)
MLH1	rs1800734 -93G>A (promoter)	GG 0× A	chip Weak	Without the risk A allele. Allele A is associated with MLH1 promoter methylation and colorectal cancer with microsatellite instability. Mechanisms: DNA Damage and DNA Repair Deficit; Epigenetic Changes
polygenic	G_MLOY Y-chromosome dosage (less mosaic loss; 14 loci)	the 86th percentile	14 variants Score	High percentile = more predicted Y dosage = less Y loss in blood cells. Y loss loci are mainly genes of DNA damage and cell cycle. Mechanisms: Somatic / Tissue mosaicism; Genomic instability; DNA Damage and DNA Repair Deficit
polygenic	PGS000662 Prostate cancer (269 variants)	the 1st percentile	221 variants Score	Very low percentile : less polygenic risk. Mechanisms: Accumulation of Somatic Mutations Genomic instability
TP53, PMF1-BGLAP, RN7SL366P, FAM117A, SPINK8	Y chromosome mosaic loss (Y measurement) 5 of 17 loci		GWAS GWAS	5 loci in which the genotype of Sérgio F deviates from the average of the reference population in the favourable sense (favourable = more Y dosage (less Y loss)). Mechanisms: Somatic / Tissue mosaicism; Genomic instability; DNA Damage and DNA Repair Deficit

5. Unfavourable

Gene/score	SNP	Genotype	Source · evidence	Effect
TP53	rs1042522 Pro72Arg	CC 2 × C	imputation (DR2=0.97) Moderate	Arg72: less cycle arrest and protective senescence; in a study in the elderly (van Heemst 2005) the carriers of Pro72 lived more despite more cancer. Mechanisms: Genomic instability; Cell Senescence; Somatic Mutation Accumulation
STARD4-AS1, NPAT	Y chromosome mosaic loss (Y measurement) 2 of 17 loci		GWAS GWAS	2 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = more Y dosage (less Y loss)). Mechanisms: Somatic / Tissue mosaicism; Genomic instability; DNA Damage and DNA Repair Deficit

6. Rare variants (ClinVar) and chip boundaries

All 35 679 variants classified as pathogenic or probably pathogenic in the ClinVar were searched for in 63 genes of the panel of this report (APEX1, ATM, ATR, BARD1, BLM, BRCA1, BRCA2, BRIP1, CHEK2, DCLRE1C, EPCAM, ERCC1, ERCC2, ERCC3, ERCC4, ERCC5, ERCC6, ERCC8, FANCA, FANCB, FANCD2, FANCE, FANCF, FANCG, FANCI, FANCM, LIG1, LIG4, MGMT, MLH1, MRE11, MSH2, MSH3, MSH6, MUTYH, NBN, NHEJ1, NTHL1, OGG1, PALB2, PARP1, PMS2, POLD1, POLE, POLH, PRKDC, RAD50, RAD51C, RAD51D, RECQL4, RNF168, SLX4, SMARCA1, TP53, WRN, XPA, XPC, XRC1, XRCC2, XRCC3, XRCC4). Of these, 4 102 have probes on the chip and additional 23 are in the imputation panel.

Gene	Variant	Genotype	Condition (ClinVar)	Evaluation
ATM	rs780905851 11:108345889 T>G	GG	Heredity cancer-predisposing syndrome; Ataxia-telangiectasia syndrome; Familial cancer of breast; not provided; ATM-report	Homozygosity for a pathogenic variant of ATM would cause ataxia-telangiectasia from childhood: incompatible with a healthy adult. Artifact of the probe (already indicated in the analysis of the raw DNA).

Chip limit: the chips genotype well common variants, but often fail in the rare ones (false positives and false negatives; Weedon 2021). The absence of pathogenic variants here does not exclude hereditary syndromes. Only clinical sequencing (panel, exome or genome) in a certified laboratory does so.

7. GWAS loci

For each study: independent loci (one SNP per region of 500 kb, the most significant, $p \leq 5 \times 10^{-8}$), with the genotype of Sérgio F and the classification against the reference population. The typical loci are in the complete table (CSV); here we show those that move away from the average.

Y chromosome mosaic loss (Y measurement)

Wright DJ et al. 2017 (67 034 men). Effect allele: more Y dosage (less Y loss) = favourable. 17 loci: 5 favourable, 2 unfavourable, 7 typical, 3 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
TP53	rs78378222	G ↓	TT	0.01	chip	Favourable
PMF1-BGLAP, PMF1	rs2736609	T ↓	CC	0.35	imputation (DR2=0.99)	Favourable
RN7SL366P - C6orf118	rs381500	C ↓	AA	0.52	imputation (DR2=0.99)	Favourable
STARD4-AS1, NREP	rs56084922	G ↓	GA	0.07	chip	Unfavourable
NPAT	rs4754301	A ↓	AA	0.61	imputation (DR2=1.00)	Unfavourable
FAM117A	rs77522818	A ↓	AT	0.94	imputation (DR2=0.98)	Favourable
SPINK8 - FBXW12	rs115854006	C ↓	CT	0.96	imputation (DR2=0.96)	Favourable

8. Map for 84 aging mechanisms

Mechanism	Favourable	Unfavourable	Interpretation
DNA Damage and DNA Repair Deficit	XRCC1 rs25487; XRCC3 rs861539; ERCC2 rs13181; ERCC2 rs1799793; XPC rs2228001; BRCA2 rs11571833; CHEK2 rs17879961; MLH1 rs1800734; G_MLOG; Y chromosome mosaic loss (Y measurement)	Y chromosome mosaic loss (Y measurement)	Favourable in excision pathways (XRCC1, ERCC2, XPC) and homologous recombination (XRCC3); OGG1/PARP1 on average.
Genomic instability	XRCC1 rs25487; XRCC3 rs861539; ERCC2 rs13181; ERCC2 rs1799793; XPC rs2228001; TP53 rs1042522; TP53 rs78378222; BRCA2 rs11571833; CHEK2 rs17879961; G_MLOG; PGS000662; Y chromosome mosaic loss (Y measurement)	TP53 rs1042522; Y chromosome mosaic loss (Y measurement)	Relative strong point: common pathways without minor repair; low predicted Y loss.
Cell Senescence	TP53 rs1042522	TP53 rs1042522	Mixed
Somatic Mutation Accumulation	TP53 rs1042522; TP53 rs78378222; PGS000662	TP53 rs1042522	Prostate at the 1st percentile; typical colorectal.
Somatic / Tissue Mosaicism	TP53 rs78378222; G_MLOG; Y chromosome mosaic loss (Y measurement)	Y chromosome mosaic loss (Y measurement)	Y loss predicted at the 86th percentile of dosage (favourable).
Clonal hematopoiesis (CHIP)	CHEK2 rs17879961	—	CHEK2 I157T absent (see the stem cell report).
Epigenetic Changes	MLH1 rs1800734	—	MLH1 –93 GG and MGMT without the methylation allele: favourable/typical.
Oxidative stress	—	—	OGG1 Ser/Being: typical repair of 8-oxoguanine.

9. Practical section — suggestion (not applied)

Coherent with the existing genetic plan and the current stack reported. Nothing was applied here; everything is to discuss with the doctor.

#	Suggestion	Why (genes)	State
1	Daily sun protection (FPS 50, clothes) and avoid burns	UV is the largest external source of damage in DNA; isotretinoin increases photosensitivity. ERCC2/XPC without minor repair alleles help, but do not eliminate the damage.	suggestion (not applied)
2	Keep the sulforaphane (already in the stack)	Induces the enzymes of II phase (NRF2) that neutralize carcinogens before forming fertilizers in DNA.	suggestion (not applied)
3	Avoid processed meat and carbonized foods; cook at lower temperatures	It reduces heterocyclic amines and aromatic hydrocarbons that form fertilizers (repaired by nucleotide excision).	suggestion (not applied)
4	More fiber (more is always better)	The butyrate protects the epithelium of the colon and reduces the damage in the DNA of the colonocytes; the colorectal score is typical.	suggestion (not applied)
5	Regular exercise and sufficient sleep	They reduce oxidative damage of DNA and improve the ability to repair.	suggestion (not applied)
6	Ferritin below 100; without supplementary iron	The free iron catalyzes radicals that oxidize DNA (8-oxoguanine).	suggestion (not applied)
7	Cancer screenings suitable for age (colon, skin) and avoiding unnecessary medical radiation	They detect early somatic mutations.	suggestion (not applied)
8	Clinical panel of hereditary cancer (BRCA1/2, Lynch, ATM, CHEK2, PALB2, MUTYH) if there is a family history or to be sure	The chip does not exclude rare variants.	suggestion (not applied)

Avoiding or with caution

- Tobacco smoke and vaping, and exposure to combustion fumes. (*suggestion, not applied*)
- Artificial tanning. (*suggestion, not applied*)

10. Limitations

- The chip does not cover most of the rare variants of hereditary cancer genes or deletions/duplications (CNV); CHEK2 1100delC is not reliable in chips.
- The functional SNPs of the repair pathways (XRCC1, XRCC3, ERCC2, XPC, OGG1) have small effects and inconsistent meta-analyses.
- The mosaic loss of the real Y is only measured in the blood (chip intensity or sequencing); here it is estimated the predisposition.
- The GWAS loci show average associations in the population; the "mapped" gene is the closest and not always the causal gene.
- The effects of isolated SNPs are small; several studies with only one population. They do not allow clinical conclusions by themselves.
- Imputation: an imputed genotype may be wrong, especially in rare variants; the imputed ones are marked.
- This report uses only genes: analysis, exams or measurements were not used.

11. Full table SNP to SNP (cured SNPs)

Gene	SNP	Effect allele (+)	Genotype	Freq.	Source	Chain	Class	Evidence
XRCC1	rs25487 Arg399Gln (base excision repair)	T (risk)	CC	0.380	chip	strand + (without conversion)	Favourable	Weak
XRCC3	rs861539 Thr241Met (homological recombination)	A (risk)	GG	0.401	chip	strand + (without conversion)	Favourable	Weak
OGG1	rs1052133 Ser326Cys (8-oxoguanine glycosylase)	G (risk)	CC	0.203	chip	palindromic: confirmed by frequency (reference population 0.20)	Typical	Moderate
ERCC2	rs13181 Lys751Gln (XPD, nucleotide excision)	G (risk)	TT	0.356	chip	strand + (without conversion)	Favourable	Weak
ERCC2	rs1799793 Asp312Asn (XPD)	T (risk)	CC	0.348	chip	strand + (without conversion)	Favourable	Weak
XPC	rs2228001 Lys939Gln (recognition of lesions)	G (risk)	TT	0.400	chip	strand + (without conversion)	Favourable	Weak
ERCC5	rs17655 Asp1104His (XPG)	C (risk)	GG	0.250	chip	palindromic: confirmed by frequency (reference population 0.25)	Typical	Weak
APEX1	rs1130409 Asp148Glu (endonuclease AP)	G (risk)	GT	0.512	chip	strand + (without conversion)	Typical	Weak
PARP1	rs1136410 Val762Ala	G (risk)	AA	0.156	chip	strand + (without conversion)	Typical	Moderate
TP53	rs1042522 Pro72Arg	C (commitment)	CC	0.722	imputation (DR2=0.97)	palindromic: confirmed by frequency (reference population 0.72)	Commitment	Moderate
MDM2	rs2279744 SNP309 T>G	G (risk)	GT	0.351	chip	strand + (without conversion)	Typical	Moderate
TP53	rs78378222 polyadenylation signal (AATAAA>AATACA)	G (risk)	TT	0.013	chip	strand + (without conversion)	Favourable	Moderate
BRCA2	rs11571833 K3326*	T (risk)	AA	0.009	chip	palindromic: confirmed by frequency (reference population 0.01)	Favourable	Moderate
CHEK2	rs17879961 I157T	G (risk)	AA	0.004	chip	strand + (without conversion)	Favourable	Moderate
MLH1	rs1800734 -93G>A (promoter)	A (risk)	GG	0.263	chip	strand + (without conversion)	Favourable	Weak
MGMT	rs16906252 promoter (methylation)	T (risk)	CC	0.066	chip	strand + (without conversion)	Typical	Weak

Notes about the typical, informative and without data: OGG1 rs1052133 (CC): The Cys326 allele has lower 8-oxoguanine repair activity (the most frequent oxidative injury of DNA), especially in oxidative stress; modest association with lung cancer. ERCC5 rs17655 (GG): The His1104 allele has small and inconsistent associations with cancer; it is included by the nucleotide excision pathway. APEX1 rs1130409 (GT): The Glu148 allele is associated in some studies with higher sensitivity to ionizing radiation; the enzymatic activity is practically normal. PARP1 rs1136410 (AA): The Ala762 allele reduces the activity of PARP1 (about 30-40% in vitro), the enzyme that signals simple chain breaks. MDM2 rs2279744 (GT): G allele increases MDM2 expression, which degrades p53, and attenuates the response to DNA damage; it is associated with earlier cancer in some populations. MGMT rs16906252 (CC): The T allele is associated with methylation of the MGMT promoter and less repair of alkylations of DNA.